Weighted Serrin Collar Reduction

This note fills in the “Input S” box from `selected-boundary-layer-theorem.md` and identifies the actual remaining sharpness obstruction.

The conclusion is:

1. the weighted D_H-to-Serrin upgrade is available in selected collars, at least in the standard oscillation form, by interpolation;
2. the specific Input S used in `selected-boundary-layer-theorem.md` is even simpler: it follows from quantitative isoperimetry through $D_I(t)$;
3. therefore the missing sharp piece is not Serrin stability. It is a boundary deficit propagation theorem that avoids the $1/\eta$ loss in the near-boundary slab average.

For clarity this note is volume-normalized: $|U| = |B_1| = \omega_n$. Constants depend only on n and on the selected-minimizer collar constants.

1 Selected-Collar Hypotheses

Assume $U = \{u > 0\}$ is a Plan 1 selected minimizer already in the graph regime. Thus there are $t_0 > 0$, $q_- > 0$, $q_+ < \infty$, and $M < \infty$ such that, for every regular $0 < t < t_0$,

$$\Sigma_t = \{u = t\}$$

is a $C^{1,\gamma}$ hypersurface with uniformly controlled charts, and

$$q_- \le |\nabla u| \le q_+ \quad \text{on } \Sigma_t, \qquad [|\nabla u|]_{C^\gamma(\Sigma_t)} \le M.$$

These are exactly the estimates supplied by Plan 1 graph entry, the Bernoulli condition, and Schauder estimates in the fixed collar.

2 Weighted Variance Gives Ordinary Variance

Let

$$f_t = |\nabla u|\big|_{\Sigma_t}, \qquad \bar{f}_t = \frac{1}{P(E_t)} \int_{\Sigma_t} f_t.$$

The level-set identity gives

$$\int_{\Sigma_t} \frac{(f_t - \bar{f}_t)^2}{f_t} = \frac{m(t)}{P(E_t)^2} D_H(t).$$

Since $q_- \le f_t \le q_+$ and $m(t), P(E_t) \simeq_n 1$ in the collar,

$$\|f_t - \bar{f}_t\|^2_{L^2(\Sigma_t)} \le C D_H(t).$$

Thus $D_H(t)$ is an ordinary L^2 Serrin defect on selected collars.

3 Interpolation to Oscillation

Let $d = n - 1$. On a uniformly $C^{1,\gamma}$ compact hypersurface Σ, if $g \in C^{\gamma}(\Sigma)$, then

$$\|g\|_{L^\infty(\Sigma)} \le C\|g\|_{L^2(\Sigma)}^{\frac{2\gamma}{2\gamma+d}}[g]_{C^\gamma(\Sigma)}^{\frac{d}{2\gamma+d}}.$$

Applying this to $g = f_t - \bar{f}_t$, and using the uniform C^γ bound, gives

$$\|f_t - \bar{f}_t\|_{L^\infty(\Sigma_t)} \le CD_H(t)^{\frac{\gamma}{2\gamma+n-1}}.$$

This is the precise bridge to Serrin-stability theorems stated in terms of oscillation of the normal derivative. It is not sharp as a power of D_H, but it is enough to show that good levels are genuine approximate Serrin domains.

4 Input S Is Already Implied by the Isoperimetric Deficit

The Input S used in `selected-boundary-layer-theorem.md` was

$$\mathcal{A}(E_t)^2 \le C\left(D_H(t) + D_I(t) + |U \setminus E_t|^2\right),$$

in normalized units. This estimate does not require Serrin stability. The quantitative isoperimetric inequality gives directly

$$\mathcal{A}(E_t)^2 \le C_n D_I(t),$$

because $m(t) \simeq_n 1$ in the selected collar. Hence Input S holds with D_H and the boundary-layer term unused:

$$\boxed{\mathcal{A}(E_t)^2 \le C_n D_I(t) \le C_n\big(D_H(t) + D_I(t) + |U \setminus E_t|^2\big).}$$

So the weighted Serrin upgrade is useful diagnostic information, but it is not the bottleneck for Fraenkel asymmetry once the level set is already regular and near the ball.

5 Boundary Traces of the Level Deficits

The selected collar gives genuine boundary traces. Write

$$q(y) = |\nabla u(y)|, \qquad y \in \partial U.$$

The normal expansion

$$\Sigma_t = \left\{y - \frac{t}{q(y)}\nu_U(y) + O(t^{1+\gamma}) : y \in \partial U\right\}$$

implies

$$P(E_t) = P(U) + O(t^\gamma), \qquad m(t) = |U| - t\int_{\partial U}\frac{1}{q} + O(t^{1+\gamma}),$$

and

$$\int_{\Sigma_t}|\nabla u| \to \int_{\partial U} q, \qquad \int_{\Sigma_t}\frac{1}{|\nabla u|} \to \int_{\partial U}\frac{1}{q}.$$

Consequently

$$D_I(t) = D_I(0) + O(t^\gamma), \qquad D_H(t) = D_H(0) + O(t^\gamma),$$

where

$$D_I(0) = P(U)^2 - c_n|U|^{2-2/n},$$

and

$$D_H(0) = \left(\int_{\partial U}\frac{1}{q}\right)\left(\int_{\partial U} q\right) - P(U)^2.$$

Thus the level-set deficits do pass to the free boundary. What is still missing is a sharp estimate of these boundary traces in terms of the torsion deficit.

6 The Tempting Boundary-Deficit Theorem Is Too Strong

The sharp boundary-layer theorem would follow from the following statement, but the statement is not true on the full nearly spherical class.

Boundary deficit propagation. For selected minimizers in the graph regime,

$$D_I(0) + D_H(0) \le C\big(E(U) - E(B_1)\big).$$

If it held, choose

$$t \le c\big(E(U) - E(B_1)\big)^{1/\gamma}$$

inside the selected collar. Then

$$D_I(t) + D_H(t) \le C\big(E(U) - E(B_1)\big),$$

and

$$|U \setminus E_t|^2 \le Ct^2 = o\big(E(U) - E(B_1)\big).$$

Combining Input S with the transfer lemma gives

$$\mathcal{A}(U)^2 \le C\big(E(U) - E(B_1)\big),$$

which is the sharp Saint-Venant/Faber-Krahn stability estimate for the selected minimizer.

The obstruction is spectral. For a spherical harmonic perturbation of degree $k \ge 2$, torsion energy is of size $k\|g_k\|_2^2$, while $D_I(0)$ and $D_H(0)$ are of size $k^2\|g_k\|_2^2$. Thus high frequencies violate the linear estimate.

7 Relation to Plan 1

The correct replacement is the selected Bernoulli law. The follow-up Plan 1 note on Bernoulli spectral closure linearizes the Bernoulli condition and finds a spectral multiplier $k-1$ on spherical harmonics. Since the α-penalty forcing is only $O(\sigma)$, all $k \ge 2$ modes are suppressed when σ is small; volume and barycenter remove $k = 0, 1$.

In boundary-deficit language:

- $D_H(0)$ measures failure of the Bernoulli datum $q = |\nabla u|$ to be constant.
- $D_I(0)$ measures the perimeter deficit of the selected free boundary.
- These defects are too strong to be bounded by torsion energy for arbitrary graphs, but selected minimizers cannot carry the high-frequency modes that cause the mismatch.

So the remaining work is a Plan 1 graph-entry plus Bernoulli spectral expansion, not a stronger Serrin theorem for rough interior levels.